

Random Forests

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Random forests are an ensemble learning method that construct a large collection of decision trees and aggregate their predictions to produce a single model. They are widely used for both regression and classification problems, particularly in settings where the relationship between predictors and the response is nonlinear or involves complex interactions. The central motivation for random forests is the reduction of variance: while individual decision trees tend to be highly variable, averaging many such trees—provided they are sufficiently decorrelated—yields a more stable and accurate predictor.

The construction of a random forest begins with decision trees, which partition the feature space into a set of disjoint regions and assign predictions based on those regions. In the regression setting, a tree partitions the predictor space into regions $R_1, \dots, R_J$ and defines a piecewise constant function of the form $\hat{f}(x) = \sum_{j=1}^J c_j \mathbf{1}(x \in R_j)$, where c_j is typically the mean of the observed responses within region R_j . These regions are chosen by recursively splitting the data to minimize the residual sum of squares. In the classification setting, predictions are instead made by majority vote within each region, and splits are chosen to reduce impurity, commonly measured by the Gini index or entropy. These criteria quantify how mixed the class labels are within a node, encouraging splits that produce more homogeneous subsets.

A random forest extends this framework by introducing randomness into the tree-building process. Specifically, each tree is trained on a bootstrap sample of the data, meaning that observations

are sampled with replacement to form a new dataset of the same size. In addition, at each split within a tree, only a random subset of the available predictors is considered. This second source of randomness is critical, as it reduces the correlation between individual trees. Without it, trees built on bootstrap samples alone tend to be similar, particularly when there are strong predictors, limiting the effectiveness of averaging.

Predictions from a random forest are obtained by aggregating the outputs of the individual trees. For regression, this is done by averaging, $\hat{f}(x) = \frac{1}{B} \sum_{b=1}^B T_b(x)$, while for classification, the predicted class is determined by majority vote across the trees. The theoretical benefit of this approach can be understood through the variance of the ensemble. If each tree has variance σ^2 and pairwise correlation ρ , then the variance of the averaged predictor is approximately $\text{Var}(\hat{f}) \approx \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$. This expression highlights that increasing the number of trees reduces variance, but the reduction is most effective when the trees are weakly correlated. The random feature selection mechanism plays a key role in achieving this decorrelation.

Although random forests are often described as making few assumptions, they do rely on several important conditions. The training observations are assumed to be independent and identically distributed, and the model implicitly assumes that the distribution of the training data matches that of future observations. Furthermore, the predictors must contain sufficient information about the response for meaningful patterns to be learned. Like other tree-based methods, random forests are also limited in their ability to extrapolate beyond the range of the observed data, as predictions are constructed from averages of observed outcomes.

An additional practical advantage of random forests is the availability of out-of-bag error estimation. Because each tree is trained on a bootstrap sample, approximately one-third of the observations are omitted from the training set for that tree. These unused observations can be used to evaluate predictive performance by averaging predictions over the subset of trees in which

the observation was not included. This provides an efficient and nearly unbiased estimate of generalization error without the need for a separate validation set.

Despite their strong empirical performance, random forests are not without limitations. Their ensemble nature makes them less interpretable than simpler models, and commonly used measures of feature importance can be difficult to interpret or biased in the presence of correlated predictors. Additionally, training a large number of trees can be computationally demanding, particularly for large datasets or high-dimensional problems. Random forests may also struggle when many predictors are irrelevant, and, as noted earlier, they do not extrapolate well beyond the range of the training data.

In summary, random forests build upon the strengths of decision trees while addressing their primary weakness—high variance—through averaging and randomization. By combining bootstrap sampling with random feature selection, they produce a flexible and robust modeling approach that performs well across a wide range of regression and classification problems.

Regression Example

Classification Example